Effects of Body Position and Loading Modality on Muscle Activity and Strength in Shoulder Presses

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Abstract & Overview

Source: NSCA Guidelines & Saeterbakken & Fimland (2013) Biomechanical Evaluation
Citation: Saeterbakken, A. H., & Fimland, M. S. (2013). Effects of body position and loading modality on muscle activity and strength in shoulder presses. Journal of Strength & Conditioning Research, 27(7), 1824-1831.

1. Biomechanics of the Overhead Shoulder Press

The overhead shoulder press is a compound multi-joint vertical pushing exercise targeting the anterior and lateral deltoids, clavicular pectoralis major, triceps brachii, and trapezius.
Key joint angles:
- Starting / Lowered Rack Phase: Dumbbells or barbell held at collarbone / clavicle height. Elbows bent at approximately 90° (DOWN_THRESHOLD ≤ 95°). Wrists stacked vertically above elbows.
- Overhead Extension / Up Phase: Arms press vertically until elbows reach full lockout at 155° to 180° overhead (UP_THRESHOLD ≥ 155°). Wrists are directly above the shoulder joint center.
- Back Alignment: Spine must maintain a neutral lordotic curve (back angle ≥ 160°).

2. Common Form Errors

• Error A: Excessive Lumbar Arching ("Excessive Arch")
  - Biomechanical Cause: User leans the upper torso backward, hyperextending the lumbar spine (back angle < 140°).
  - Root Causes: Weak abdominal stabilizers, insufficient thoracic spine mobility, or attempting to turn the vertical press into an incline bench press using the larger sternal pectoralis major.
  - Risk: High facet joint compression, disc shear strain, and lower back injury.
  - Correction: Squeeze glutes firmly, pull the ribcage down, and tuck the pelvis. If shoulder mobility is limited, reduce the load or use a high-incline seated variation.
• Error B: Incomplete Overhead Lockout ("Incomplete Extension")
  - Biomechanical Cause: Failing to press past 130°-140°, reversing direction before full vertical extension.
  - Correction: Press the dumbbells straight upward until upper arms are aligned beside the ears. Ensure wrists stay stacked over elbows.
• Error C: Flaring Elbows Outward Beyond the Scapular Plane
  - Correction: Keep elbows slightly forward in the scapular plane (roughly 30° forward of the coronal plane) to reduce impingement on the supraspinatus tendon.

3. Recommended Camera Setup for Shoulder Press

- Angle: Front-facing (frontal) or slight 3/4 oblique view.
- Distance: 2.0 to 3.0 meters away.
- Overhead Framing Clearance: Camera must be tilted high enough so that when arms are fully locked overhead, the hands and wrists do not clip out of the top edge of the frame.